

SOUNIC AKKARAJU

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SUMMARY

Master's student in Computer Science (specializing in Data Science & Technology) at University of Twente, with a strong foundation in AI, Machine Learning, Deep Learning, and Big Data technologies. Experienced in backend development and full ML pipelines, with previous internship at CoreTek Labs and hands-on project work in NLP, healthcare prediction, and transformer-based model evaluation.

WORK EXPERIENCE

Software Engineering Intern, CORETEK LABS

Jul 2023 - Sept 2023

- Spearheaded the development of an AI chatbot, leveraging Django REST framework to create a seamless backend infrastructure.
 - Acquired expertise in web scraping techniques and utilized the scraped data to train the chatbot, enhancing its knowledge base for the company's website.
 - Worked closely with the back-end development team, fostering a collaborative environment that facilitated knowledge sharing and problem-solving.
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PROJECTS

Bachelor Thesis

Dec 2024 - May 2025

Miscarriage Prediction Using Ensemble Deep Learning Models

- Developed novel NearSMOTE class-balancing technique combining SMOTE, NearMiss, and Tomek links to handle severe class imbalance in healthcare prediction tasks.
 - Built ensemble deep learning system combining TabTransformer, FT-Transformer, and TabNet architectures, achieving 93.1% AUC-ROC on NFHS-5 national health survey data.
 - Implemented end-to-end ML pipeline: LASSO feature selection, class balancing, model training (PyTorch on A100 GPU), and SHAP-based interpretability analysis.
 - Identified actionable clinical insights through SHAP analysis — prenatal care and health checks as top protective factors against pregnancy complications
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Natural Language Processing Project

Sept 2025 - Nov 2025

Exploring In-Context Learning and LoRA vs BERT

- Collaborated on 2-person research project comparing parameter-efficient fine-tuning approaches across classification, QA, and reasoning tasks using 10 benchmark datasets.
- Implemented BERT/DistilBERT fine-tuning pipeline achieving 93.7% accuracy on AG News; analyzed cross-task transfer limitations (near-random performance on out-of-domain tasks)
- Conducted systematic in-context learning experiments on Gemma3 models (270M–4B parameters), evaluating k-shot performance (k=0,5,10,25) across multiple decoding strategies.
- Quantified efficiency tradeoffs: BERT delivers 190x higher throughput (3837 vs 20 tokens/sec) but zero cross-task transfer; ICL context expansion increases latency 26-40%
- Co-authored research article documenting deployment recommendations for single-task vs multi-task production systems

Automated Chore Management Mobile App

- Built a cross-platform mobile app (Flutter/Dart) with Supabase backend serving PostgreSQL database with Row-Level Security, real-time subscriptions, and Edge Functions.
- Integrated Google Calendar API with OAuth 2.0 for automatic chore syncing, handling token refresh and calendar event CRUD operations.
- Designed and implemented 7-table relational database schema supporting households, role-based permissions, recurring chores, and user preference-based scheduling.
- Architected app using feature-first structure with Repository pattern, separating UI, business logic, and data layers across 15+ feature modules.

Managing Big Data

Dec 2025 - Feb 2026

Crypto Market Microstructure Analysis

- Built distributed data pipeline using PySpark on university HDFS cluster to process Binance full trading history across 1000+ cryptocurrency pairs.
- Engineered minute-level features: log returns, Amihud illiquidity, Parkinson volatility, taker ratios, and zero-volume flags from raw trade-level data.
- Developed 3 analytical Spark jobs: stress vs non-stress liquidity comparison, BTC/ETH shock propagation lag analysis (0–120 min), and regime-dependent sensitivity estimation.
- Implemented efficient processing with adaptive query execution, partition strategies, and Parquet columnar storage for scalable cluster deployment.

EDUCATION**MSc Computer Science (specialization in Data Science & Technologies)**

Sep 2025 - Jun 2027

University Of Twente, Netherlands

- Core Modules: Machine Learning, Deep Learning, Natural Language Processing, Big Data

BSc Computer Science and Artificial Intelligence

Sep 2023 - Jun 2025

Aberystwyth University, United Kingdom

- Core Modules: Scientific Python (1st), Software Engineering (1st), Modelling Persistent Data (1st)

TECHNICAL SKILLS

- **Programming Languages:** Python, Dart, JavaScript/TypeScript, Java, SQL, HTML/CSS.
- **ML/AI:** PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, SHAP, Pandas, NumPy, Im-eval.
- **Frameworks:** Flutter, Django REST, React, Next.js, PostgreSQL, Supabase, Git
- **Cloud & APIs:** Google Cloud (OAuth 2.0, Calendar API), Supabase (Auth, Storage, Realtime, Edge Functions)
- **Concepts:** Deep Learning, NLP, Transformers, Parameter-Efficient Fine-Tuning, REST APIs, Database Design.

LANGUAGES

- **English (Fluent) • Hindi (Fluent) • Telugu (Native) • Dutch (Learning)**